

CTU-Chat: Supervisor-Routed Multi-Agent RAG with Heterogeneous Knowledge Allocation for University Counseling

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Abstract. Higher education student counseling involves heterogeneous institutional knowledge, spanning relational curriculum graphs, cohort-specific tuition matrices, statutory exemption criteria, and narrative administrative decrees. Monolithic RAG frameworks and uncoordinated agent meshes can be brittle in this setting because of uniform representations, cross-domain evidence interference, and unconstrained tool exposure. We present **CTU-Chat**, a centralized supervisor-routed, domain-specialized multi-agent architecture in which each specialist is assigned a domain-specific knowledge space, retrieval path, and tool privilege for Vietnamese university counseling. Under centralized supervisor coordination, CTU-CHAT dispatches user queries to four bounded specialists with asymmetric privileges across three tiers: relational academic curricula are resolved through dedicated Cypher tools over a Neo4j knowledge graph, structured tuition rates are resolved via graph lookup and JSON catalogs with deterministic arithmetic engines, and narrative administrative regulations are retrieved through a hybrid document RAG path (BM25 and dense embeddings fused via Reciprocal Rank Fusion and cross-encoder reranking). Multi-agent evaluation demonstrates 97.0% supervisor routing accuracy (97.8% Macro-F1) and 100.0% bounded completion under strict invariants (`max_tool_calls` = 1, `timeout` = 60.0 s). Across 60 production-tool cases repeated three times, academic lookup, tuition-policy, scholarship, and no-tool gates achieved 100.0% end-to-end pass, while graph tuition lookup achieved 80.0% and graph exemption-basis lookup 73.3%. On the 150-question benchmark, progressive component stacking achieves 0.9000 Hit@1, 0.9600 Hit@3, 0.8656 Recall@5, and 0.9327 MRR.

Keywords: Multi-Agent Systems · Domain-Specialized Agents · Centralized Supervisor Routing · Heterogeneous Knowledge Allocation · Retrieval-Augmented Generation · University Counseling

1 Introduction

Providing timely, accessible, and accurate administrative guidance is vital for student success and institutional efficiency. In large universities such as Can Tho University (CTU)—a flagship institution in Vietnam’s Mekong Delta serving over 40,000 students [1]—advisory inquiries span diverse administrative domains. Because regulations are frequently updated through ministerial circulars, rector decisions, and cohort-specific guidelines, automated conversational counseling systems are essential to alleviate advisory strain and ensure compliance.

However, higher education counseling is characterized by acute *knowledge heterogeneity*, encompassing structurally distinct representations: (1) **Academic Curricula**: relational course prerequisite chains spanning the 113 majors represented in our graph corpus; (2) **Tuition Schedules**: multi-dimensional rate matrices governed by admission cohort and discipline; (3) **Statutory Exemptions**: legal decrees prescribing percentage deductions for qualified beneficiaries [6,7]; (4) **Scholarships**: multi-variable eligibility thresholds (cumulative GPA and training point scores); and (5) **Campus Regulations**: narrative administrative prose governing examinations, conduct, and student loans.

To automate institutional QA, Retrieval-Augmented Generation (RAG) [10,5] and GraphRAG [3] augment large language models (LLMs) with external documents or knowledge graphs, while agentic architectures [20,18,16] introduce iterative tool invocation and multi-agent coordination. Nevertheless, applying existing paradigms to university counseling reveals significant limitations. Monolithic RAG pipelines risk a *uniform representation fallacy*: forcing relational curricula, tabular fee matrices, and narrative legal prose into a single retrieval index can fragment tabular data and prerequisite links. Conversely, uncoordinated multi-agent meshes can expose broad tool sets across agents, increasing the risk of tool mis-selection, parameter errors, and extended execution, while known limitations in language-model arithmetic [14] motivate deterministic calculation for credit-based tuition.

These challenges crystallize three open research gaps in automated university counseling: (1) **Gap 1 (Limited Domain-Specialized Multi-Agent Orchestration)**: Existing agentic and multi-agent systems provide delegation and tool use, but the coordinated use of domain-specialized agents with isolated knowledge spaces, tool privileges, and retrieval paths remains underexplored in heterogeneous university counseling; (2) **Gap 2 (Uniform Knowledge Representation)**: Existing RAG-oriented systems often process structurally different institutional knowledge through a common retrieval space, although relational curricula, structured tuition information, and narrative regulations require fundamentally different representations; and (3) **Gap 3 (Lack of Representation-Aware Routing)**: Existing systems provide lexical, semantic, and graph retrieval methods, but offer limited mechanisms for selecting the appropriate evidence representation and specialist according to query domain and information structure.

To address these gaps, we formulate three core research questions:

- **RQ1 (Domain-Specialized Multi-Agent Architecture):** How effectively can a centralized supervisor-routed multi-agent architecture coordinate domain-specialized agents with isolated knowledge and tool access for heterogeneous university counseling tasks?
- **RQ2 (Heterogeneous Knowledge Allocation):** How does prioritizing graph-oriented access for relational academic-program and structured tuition knowledge, while retaining narrative regulations in hybrid document retrieval, affect retrieval and answer quality compared with uniform retrieval representations?
- **RQ3 (Representation-Aware Evidence Routing):** How does intent-guided routing between graph-oriented and hybrid document-retrieval paths affect retrieval effectiveness, answer correctness, and latency across different counseling domains?

We hypothesize that reliable university counseling is better supported by a supervisor-routed multi-agent architecture in which domain specialists are constrained to knowledge representations and tools appropriate to their information structure, rather than by a uniform retrieval-and-reasoning pipeline. Accordingly, CTU-CHAT separates specialist coordination, knowledge allocation, and evidence-path selection as three complementary design dimensions.

To answer these questions, we present **CTU-Chat**. CTU-CHAT *proposes a centralized supervisor-routed, domain-specialized multi-agent architecture in which each specialist is assigned a domain-specific knowledge space, retrieval path, and tool privilege*. Under centralized supervisor coordination, incoming inquiries are contextualized via Redis history and dispatched to four bounded specialists operating under asymmetric privileges: (1) **Academic Specialist**, bound to 6 dedicated Cypher tools over a Neo4j knowledge graph; (2) **Financial Specialist**, equipped with structured tuition graph lookups and the deterministic arithmetic engine `tin_h_toan_hoc_phi`; (3) **Scholarship Specialist**, equipped with the deterministic evaluator `tin_h_tien_hoc_bong`; and (4) **General Specialist**, operating with zero tools to synthesize narrative regulatory context. Specialists share session state in Redis while maintaining strictly isolated execution envelopes without unconstrained peer-to-peer communication.

Our principal contributions are three-fold:

- **C1 (Supervisor-Routed Domain-Specialized Multi-Agent Architecture):** A centralized supervisor-routed architecture coordinating four domain-specialized agents with asymmetric tool privileges, a per-decision bound of `max_tool_calls = 1`, and shared session state, designed to reduce tool mis-selection and constrain execution without peer-to-peer delegation (addressing Gap 1 and RQ1).
- **C2 (Heterogeneous Knowledge Allocation Framework):** A principled allocation strategy that separates relational curricula and structured tuition into graph-oriented representations while reserving hybrid document retrieval for narrative administrative decrees, resolving the uniform representation fallacy (addressing Gap 2 and RQ2).

- **C3 (Representation-Aware Routing and Evidence Acquisition):** An intent-guided evidence acquisition mechanism that routes queries dynamically to graph-oriented or hybrid document-retrieval paths, leveraging established BM25, Dense vectors, RRF, and cross-encoder reranking as complementary building blocks (addressing Gap 3 and RQ3).

2 Related Work

This section reviews established retrieval and reasoning paradigms in the literature, examining their operational mechanisms and limitations in institutional counseling.

2.1 Sparse and Dense Retrieval

Lexical Retrieval (BM25): Sparse keyword search using BM25 [15] evaluates term frequency (tf) and inverse document frequency (IDF) over an inverted index:

$$s_{\text{BM25}}(q, d) = \sum_{w \in q} \text{IDF}(w) \cdot \frac{\text{tf}(w, d) \cdot (k_1 + 1)}{\text{tf}(w, d) + k_1 \cdot \left(1 - b + b \cdot \frac{|d|}{|d|}\right)}, \quad (1)$$

with $k_1 = 1.5$ and $b = 0.75$. BM25 supports exact lexical matching for rigid alphanumeric tokens (decision codes, cohort labels, course identifiers), but may miss semantic paraphrases.

Dense Semantic Retrieval: Dense retrieval systems [9,10] map queries and passages into embedding spaces using bi-encoders, computing relevance via vector similarity. While dense models capture paraphrastic intent, rare alphanumeric identifiers and out-of-vocabulary regulatory tokens can remain difficult and may introduce cross-cohort noise, motivating hybrid retrieval.

2.2 Hybrid Retrieval and Reciprocal Rank Fusion

To merge BM25 and dense candidates without score calibration, Reciprocal Rank Fusion (RRF) [2] sorts documents by reciprocal position: $s_{\text{RRF}}(d) = \sum_{s \in \mathcal{M}} \frac{1}{k + r_s(d)}$ ($k = 60$), while neural cross-encoders [12,19] jointly encode query–passage pairs for reranking. Adaptive-RAG further demonstrates that retrieval strategies can be selected according to question complexity [8]. These approaches motivate dynamic retrieval, but do not directly allocate institution-specific relational, tabular, and narrative knowledge to isolated specialist paths.

2.3 Knowledge Graphs and GraphRAG

Knowledge graphs represent domain knowledge as structured triples (h, r, t) [13]. GraphRAG [3] leverages graph topology to uncover multi-hop associative paths escaping flat chunking, benefiting curriculum structures and prerequisite chains. However, universal graph extraction across institutional corpora produces hallucinated relationships, while forcing narrative legal prose into graphs is structurally unnatural, motivating selective rather than universal graph use.

2.4 Agentic RAG and Multi-Agent Orchestration

ReAct demonstrates interleaved reasoning and action, AutoGen provides a framework for multi-agent conversation, and Toolformer studies when and how language models invoke external APIs [20,18,16]. For institutional counseling, these capabilities raise design risks when applied without explicit constraints: broad tool exposure can increase tool and parameter errors; peer-to-peer exchanges can increase coordination cost; a shared index can mix evidence across domains; and a common retrieval path can obscure opportunities for direct graph traversal or structured lookup. These risks motivate the bounded, supervisor-routed specialization evaluated in this work.

2.5 Educational and University Counseling Systems

Systematic evidence shows that educational chatbots are used for informational, assisting, mentoring, and learning-support roles, while evaluation quality and adaptation remain open challenges [17]. REBot is a recent example of applying semantically enriched RAG and graph routing to educational counseling [11]. However, this line of work does not establish that a single retrieval representation or generative arithmetic is appropriate for every counseling domain. We therefore study explicit representation allocation, specialist routing, and deterministic calculation as separate design dimensions.

2.6 Comparative Synthesis and Research Gaps

Synthesizing established paradigms reveals distinct structural trade-offs for our setting: (1) *BM25* [15] favors exact administrative tokens but may miss natural paraphrases; (2) *Dense Retrieval* [9] captures semantic intent but may confuse similar cohort-specific passages; (3) *Hybrid RRF* [2] combines lexical and semantic rankings but does not itself model relational curricula; (4) *GraphRAG* [3] supports graph-based associations, although universal graph extraction is not necessarily suitable for narrative regulations; (5) *Adaptive RAG* [8] selects retrieval depth by query complexity but not by institutional representation type; (6) *ReAct/Toolformer-style systems* [20,16] support tool-using reasoning but require tool-access controls; and (7) *multi-agent conversation frameworks* [18] provide coordination primitives but do not by themselves guarantee bounded execution or knowledge isolation.

This comparative analysis directly crystallizes the three core research gaps identified in Section 1: (1) **Gap 1 (Limited Domain-Specialized Multi-Agent Orchestration)**: Coordinated multi-agent systems with isolated knowledge spaces, tool privileges, and dedicated retrieval paths remain underexplored in university counseling; (2) **Gap 2 (Uniform Knowledge Representation)**: Processing relational curricula, structured tuition, and narrative regulations through a common retrieval space induces representation failure; and (3) **Gap 3 (Lack of Representation-Aware Routing)**: Dynamic routing mechanisms that steer queries to appropriate evidence representations and bounded specialists according to information structure are absent in existing architectures.

3 Proposed Model

This section presents the architectural design, algorithmic foundation, and operational mechanisms of CTU-CHAT. Unlike conventional monolithic RAG pipelines that force all institutional documentation into a uniform document index or graph schema, CTU-CHAT *proposes a centralized supervisor-routed, domain-specialized multi-agent architecture in which each specialist is assigned a domain-specific knowledge space, retrieval path, and tool privilege.*

3.1 Design Motivation and Knowledge Characteristics

Higher education counseling at large institutions such as Can Tho University (CTU) is characterized by acute knowledge heterogeneity. An empirical audit of 244 official institutional documents reveals five structurally distinct categories of administrative information, each imposing unique representation and reasoning requirements:

1. **Academic Curricula (Relational Knowledge):** Encompassing 113 undergraduate majors (alongside an institutional academic regulation document), curricula exhibit deep hierarchies (General → Fundamental → Specialized → Electives) and strict multi-hop prerequisite dependencies. While flat document chunking fragments prerequisite chains, a property graph naturally captures topological relationships, enabling deterministic path traversals without relation hallucination.
2. **Tuition Fee Schedules (Structured / Parametric Knowledge):** Governed by official institutional rector decisions, tuition fee schedules are multi-dimensional matrices indexed by admission cohorts and major groups. Flat text search frequently conflates cohorts, while generative LLMs miscalculate credit multipliers, requiring structured tabular lookups coupled with deterministic arithmetic engines.
3. **Statutory Fee Reductions (Normative Legal Decrees):** Governed by Decree 81/2021/ND-CP and its amendment, Decree 97/2023/ND-CP [6,7], statutory exemptions prescribe exact deductions for qualified beneficiaries. These require parent-child hybrid text retrieval for legal qualification, followed by exact mathematical deduction against student tuition.
4. **Scholarships and Financial Aid (Rule-Based Threshold Policies):** Academic encouragement scholarships operate under dual thresholds (cumulative GPA and training point scores) evaluated across tiers (Good, Very Good, Excellent) against semester quotas, combining rule-based filtering with narrative conditions.
5. **General Campus Regulations (Unstructured Administrative Prose):** Administrative decisions governing student conduct, examination retakes, dormitories, and deferrals are long narrative prose requiring high-recall semantic search paired with exact lexical matching for official decision identifiers and administrative codes.

These heterogeneous characteristics expose the fundamental flaw of uniform RAG systems: no single retrieval or reasoning paradigm is optimal for all administrative domains. This motivates our proposed architecture: coordinating domain-specialized agents with isolated knowledge spaces and asymmetric tool privileges under a centralized supervisor.

3.2 Overall Proposed Architecture

Figure 1 depicts the system architecture of CTU-CHAT, structured across three coordinated tiers: (1) the *Interaction and Centralized Supervisor Tier*, (2) the *Domain-Specialized Multi-Agent Core Layer*, and (3) the *Heterogeneous Knowledge Storage and Tool Tier*.

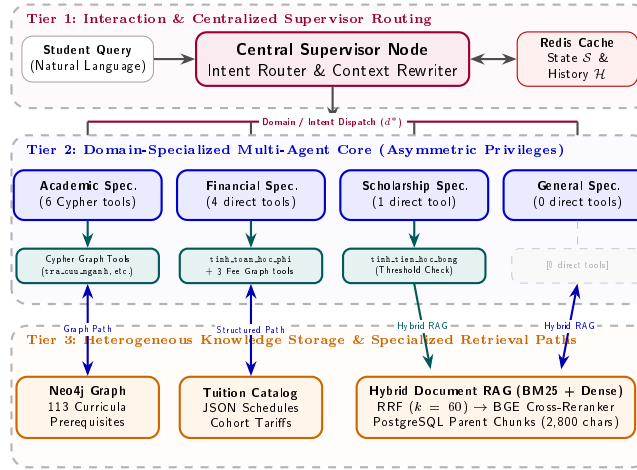


Fig. 1. System architecture of CTU-CHAT, highlighting the centralized supervisor, the domain-specialized multi-agent core layer, and separate knowledge paths for relational, structured, and narrative administrative evidence.

The interaction flow proceeds through three well-defined stages: (1) **Tier 1 (Supervisor)**: Student query q is contextualized against Redis history $\mathcal{H}$ and routed to predicted domain $d^* \in \mathcal{D}$; (2) **Tier 2 (Domain Multi-Agent Core)**: The query is dispatched to the designated specialist operating under asymmetric tool privileges (Table 1); and (3) **Tier 3 (Heterogeneous Knowledge Paths)**: The specialist queries isolated representations (Neo4j for curricula, structured graph/catalogs for fee matrices, and hybrid document RAG for narrative regulations) and executes bounded deterministic tools.

3.3 Supervisor-Routed Domain-Specialized Multi-Agent Architecture

A core contribution of CTU-CHAT is the operational isolation of domain agents within a **centralized supervisor-routed multi-agent architecture**. In conventional multi-agent meshes, exposing broad tool suites to all agents and permitting autonomous agent-to-agent negotiation can increase the risk of parameter

errors, cascading failures, and extended execution. CTU-CHAT is designed to reduce these risks through explicit structural boundaries:

- **Supervisor Agent (Central Dispatcher):** Implemented via a structured output LLM (`gemini-2.5-flash-lite` in empirical evaluation, configurable to `gemini-3.5-flash-lite` in operational deployments) with deterministic regex fallbacks. The supervisor inspects incoming query q and session history $\mathcal{H}$, generates a disambiguated standalone query $q_{\text{standalone}}$, classifies intent into exactly one target domain $d^* \in \{\text{academic, financial, scholarship, general}\}$, and selects the appropriate evidence acquisition path.
- **Four Domain Specialists:**
 1. **Academic Specialist:** Dedicated to curriculum structures across 113 undergraduate majors, prerequisite course dependency chains, and course equivalence. Bounded to 6 direct Cypher inspection tools connecting to Neo4j, with zero arithmetic tools and zero document retrieval tools. Reasoning mode: ReAct loop over graph tools.
 2. **Financial Specialist:** Dedicated to credit-based tuition fee schedules, cohort tariffs, statutory fee deductions, and payment deadlines. Armed with 4 direct tools including fee graph queries and the deterministic arithmetic engine `tin_hoc_phi`.
 3. **Scholarship Specialist:** Dedicated to academic encouragement scholarships, corporate sponsored grants, and financial aid quotas. Armed with the deterministic threshold evaluation tool `tin_hoc_bong`.
 4. **General Specialist:** Dedicated to administrative regulations, student conduct, examination retakes, dormitories, student loans, and pedagogical support. Operates with zero direct tools, synthesizing answers purely over retrieved hybrid parent-child regulatory passages.
- **Shared State vs. Isolated Resources:** All agents share conversational session state $\mathcal{S}$ and dialogue history $\mathcal{H}$ managed in Redis. However, tool namespaces, prompt envelopes, and knowledge stores are strictly isolated: no specialist has access to cross-domain tools or unauthorized databases.
- **Strict Centralized Routing (No Peer-to-Peer Communication):** Unlike collaborative agent meshes, CTU-CHAT strictly prohibits peer-to-peer handoffs, autonomous agent-to-agent negotiations, or dynamic delegation. All interactions follow the centralized routing topology: $\text{START} \rightarrow \text{Supervisor} \rightarrow [\text{Evidence Node}] \rightarrow \text{Specialist} \rightarrow \text{END}$.

Table 1. Operational comparison and resource privileges of domain-specialized agents in CTU-CHAT.

Specialist Agent	Shared State	Available Direct Tools	Primary Evidence Store	Retrieval Strategy	Tool-Call Bound
Academic Specialist	Redis $\mathcal{S}, \mathcal{H}$	6 Cypher tools (<code>tra_cus_nhanh</code> , etc.)	Neo4j Knowledge Graph	Direct Cypher topological traversal	<code>max_tool_calls = 1</code>
Financial Specialist	Redis $\mathcal{S}, \mathcal{H}$	4 tools (<code>tin_hoc_phi</code> , etc.)	Tuition Graph + JSON Catalog	Pre-retrieved Fee Graph/JSON + Calc	<code>max_tool_calls = 1</code>
Scholarship Specialist	Redis $\mathcal{S}, \mathcal{H}$	1 tool (<code>tin_hoc_bong</code>)	Hybrid RAG + Criteria Catalog	Pre-retrieved Hybrid + Criteria tool	<code>max_tool_calls = 1</code>
General Specialist	Redis $\mathcal{S}, \mathcal{H}$	0 direct tools (pure synthesis)	PostgreSQL + Qdrant	Parent-child Hybrid RRF + BGE	<code>max_tool_calls = 0</code>

Distinction Between Direct Tools and Pre-Retrieved Context: Table 1 formalizes the boundary between *direct tools bound to specialists* and

pre-retrieval functions executed outside the specialist. Specifically, the **Academic Specialist** executes its 6 Cypher tools directly during ReAct execution without passing through the text retrieval node. In contrast, for the **Financial Specialist** and **Scholarship Specialist**, structured tuition tables and hybrid regulatory passages are pre-retrieved and formatted into prompt context by the upstream retrieval node; direct tools are then invoked solely for deterministic calculations or interactive sub-queries.

3.4 Heterogeneous Knowledge Allocation

Rather than forcing all university data into a single retrieval index, CTU-CHAT allocates knowledge across specialized modalities according to structural properties (Table 2).

Table 2. Heterogeneous knowledge allocation across institutional administrative domains.

Knowledge Type	Structural Property	Primary Representation	Runtime Access Mechanism	Specialist
Academic Programs	Relational, multi-hop	Neo4j Knowledge Graph	Dedicated Cypher query tools	Academic
Tuition Schedules	Structured, cohort-dependent	Neo4j Tuition Graph / JSON	Graph lookup + deterministic calc	Financial
Scholarship Criteria	Narrative + rule-based thresholds	Hybrid Document RAG	Hybrid search + threshold tool	Scholarship
General Regulations	Narrative administrative text	Hybrid Document RAG	BM25 + Dense + RRF + BGE	General

Physical Storage vs. Primary Runtime Retrieval Path: We draw a crucial architectural distinction between physical indexing and runtime retrieval:

- *Physical Storage/Indexing:* Neo4j 5 hosts topological curriculum entities (**Major**, **KnowledgeBlock**, **Course**) and structured tuition nodes. Structured JSON files store frozen credit tariffs and scholarship thresholds. Simultaneously, 244 institutional regulatory decrees are archived in PostgreSQL (parent chunks, 2,800 characters) and Qdrant (child chunks, 400 characters).
- *Primary Runtime Retrieval Path:* At runtime, academic inquiries bypass text search entirely to query Neo4j directly via Cypher tools. For tuition inquiries, the primary runtime path executes structured graph/catalog lookup; document retrieval is only invoked for textual decrees (e.g., payment procedures). For scholarship and general queries, the runtime path executes hybrid document retrieval.

3.5 Representation-Aware Routing Strategy

Applying global hybrid retrieval across an undifferentiated administrative corpus introduces severe cross-domain evidence interference. To ensure precision, CTU-CHAT enforces rigorous decision criteria mapping inquiry characteristics to preferred evidence mechanisms (Table 3).

Formally, the supervisor maps incoming student query q and dialogue context $\mathcal{H}$ to a constrained routing tuple:

$$\mathcal{R}(q, \mathcal{H}) = \langle q_{\text{standalone}}, d^*, A^*, \mathcal{P}_{A^*}, \mathcal{L}_{\text{target}} \rangle, \quad (2)$$

where $q_{\text{standalone}}$ is the disambiguated query, $d^* \in \mathcal{D}$ denotes the predicted domain ($\mathcal{D} = \{\text{acad}, \text{fin}, \text{sch}, \text{gen}\}$), A^* is the assigned specialist agent, $\mathcal{P}_{A^*}$ defines

Table 3. Decision criteria for evidence representation and retrieval mechanism selection.

Information Characteristic	Preferred Mechanism	Structural Rationale
Exact identifiers, course codes, formal terms	BM25 Lexical Search	Exact inverted index token matching without semantic drift
Semantic paraphrase & conversational phrasing	Dense Vector Search (Qdrant)	Captures contextual proximity in 768-d embedding space
Narrative administrative policies & decrees	Hybrid BM25 + Dense + RRF + BGE	Fuses lexical and semantic signals, refined by cross-attention
Relational curriculum / prerequisite query	Neo4j Knowledge Graph	Deterministic graph traversal over topological dependencies
Statutory tuition / scholarship calculation	Structured Graph/JSON + Tools	Deterministic arithmetic outside generative text production

authorized tool privileges, and $\mathcal{L}_{\text{target}} \in \{\text{'direct_graph'}, \text{'evidence_node'}\}$ specifies the evidence acquisition path.

After identifying the need for contextualization, the supervisor jointly determines the target specialist and its corresponding evidence path. Algorithm 1 formalizes this centralized routing process. Given the current student query q , dialogue history $\mathcal{H}$, and shared session state $\mathcal{S}$, the supervisor optionally rewrites context-dependent queries, classifies domain and intent, assigns a bounded specialist, and selects the appropriate graph-oriented or document-oriented evidence path.

Algorithm 1: Supervisor Intent and Specialist Routing

Input: Incoming student query q , Session dialogue history $\mathcal{H}$, Shared session state $\mathcal{S}$
Output: Routing tuple $\langle q_{\text{standalone}}, d^*, A^*, \mathcal{P}_{A^*}, \mathcal{L}_{\text{target}} \rangle$

```

1  $q_{\text{standalone}} \leftarrow q$ ;
2 if ShouldRewriteQuery( $q, \mathcal{H}$ ) then
3    $q_{\text{cand}} \leftarrow \text{FastLLM\_Rewrite}(q, \mathcal{H})$ ;
4   if ValidateRewrite( $q, q_{\text{cand}}, \mathcal{H}$ ) then
5      $q_{\text{standalone}} \leftarrow q_{\text{cand}}$ ;
6  $\langle d^*, \iota \rangle \leftarrow \text{ClassifyDomainAndIntent}(q_{\text{standalone}}, \text{SUPERVISOR\_PROMPT})$ ;
7 if  $d^* = \text{NULL}$  then
8    $d^* \leftarrow \text{RegexFallbackClassifier}(q_{\text{standalone}})$ ;
9  $A^* \leftarrow \text{SelectSpecialist}(d^*)$ ;  $\mathcal{P}_{A^*} \leftarrow \text{GetSpecialistPrivileges}(A^*)$ ;
10 if  $d^* = \text{'academic'}$  then
11    $\mathcal{L}_{\text{target}} \leftarrow \text{'direct\_graph'}$ ;
12 else
13    $\mathcal{L}_{\text{target}} \leftarrow \text{'evidence\_node'}$ ;
14 return  $\langle q_{\text{standalone}}, d^*, A^*, \mathcal{P}_{A^*}, \mathcal{L}_{\text{target}} \rangle$ ;

```

The output of Algorithm 1 is not the final answer but a constrained routing decision tuple. This decision limits downstream execution strictly to the selected specialist A^* , designated evidence source $\mathcal{L}_{\text{target}}$, and permitted tool set $\mathcal{P}_{A^*}$, thereby enforcing the resource isolation that defines the proposed multi-agent architecture. By pruning unauthorized tools prior to specialist invocation, this centralized routing mechanism constrains tool exposure and execution paths; Scenario 3 evaluates the resulting bounded behavior at the isolated decision level.

3.6 Hybrid Document Retrieval Path

For narrative administrative regulations and scholarship policies, CTU-CHAT employs a multi-stage hybrid document retrieval pipeline:

1. **Dual Candidate Generation:** Query $q_{\text{standalone}}$ is searched simultaneously via BM25 over tokenized child chunks in PostgreSQL and dense vector search

in Qdrant (768-d embeddings via `vietnamese-bi-encoder`), returning candidate sets $\mathcal{D}_{\text{bm25}}$ and $\mathcal{D}_{\text{dense}}$ (top-15 in production defaults, with $k = 10$ evaluated in retrieval benchmarks).

2. **Reciprocal Rank Fusion (RRF)**: Candidates are merged without fragile score normalization:

$$s_{\text{RRF}}(d) = \sum_{s \in \{\text{bm25}, \text{dense}\}} \frac{1}{k + r_s(d)}, \quad k = 60. \quad (3)$$

3. **Neural Cross-Encoder Reranking**: The top-fused candidates are reranked via `BAAI/bge-reranker-v2-m3` cross-attention, selecting the top-5 child chunks (with top-6 configured as the engine production ceiling).
4. **Parent Chunk Expansion**: Selected child chunks are mapped back to their 2,800-character parent chunks in PostgreSQL, ensuring complete legal context and article structure for generation.

Clarification of Component Novelty: We emphasize that BM25, dense bi-encoders, RRF, and cross-encoders are established literature paradigms. In CTU-CHAT, they function as building blocks within the hybrid path, coordinated under supervisor routing.

3.7 Graph-Oriented Academic and Tuition Retrieval

Curriculum Knowledge Graph (Neo4j): Curricula across all 113 undergraduate programs are modeled in Neo4j (alongside general academic regulations). The **Academic Specialist** queries this graph directly via 6 APOC-optimized Cypher tools, resolving complex multi-hop prerequisite chains deterministically without relation hallucination.

Tuition Knowledge Graph and Structured Catalogs: Credit-based tuition fee schedules are indexed as structured nodes in Neo4j with a JSON catalog fallback. For statutory calculations, parameter extraction invokes the deterministic engine `tinhh.toan.hoc.ph`:

$$\text{Gross} = \sum_{i \in \mathcal{C}} c_i \cdot r_i \cdot m_i, \quad (4)$$

$$\text{Reduction} = \sum_{i \in \mathcal{C}} c_i \cdot b_i \cdot (p_i/100), \quad (5)$$

$$\text{Payable} = \max(0, \text{Round}_{\text{VND}}(\text{Gross} - \text{Reduction})), \quad (6)$$

where $\mathcal{C}$ is the registered course set, c_i is course credit count, r_i is per-credit fee, m_i is cohort multiplier, b_i is exemption base rate, and $p_i \in \{0, 50, 70, 100\}$ is statutory deduction percentage. A typed Pydantic tool schema ensures numerical reliability.

3.8 Domain-Aware Evidence Acquisition and Specialist Execution

Algorithm 2 executes the routing decision produced by Algorithm 1. Evidence acquisition is intentionally asymmetric across specialists: academic queries are resolved through graph-oriented evidence, financial queries use structured tuition

evidence and deterministic calculation when required, while narrative administrative queries are served through hybrid document retrieval. This separation is central to CTU-CHAT and distinguishes it from a uniform retrieve-then-generate pipeline.

Algorithm 2: Evidence Acquisition and Bounded Specialist Execution

Input: Routing tuple $\langle q_{\text{standalone}}, d^*, A^*, \mathcal{P}_{A^*}, \mathcal{L}_{\text{target}} \rangle$, Knowledge stores (Neo4j, Qdrant, PostgreSQL, JSON)

Output: Validated citation-grounded response y_{final}

```

1 if  $A^* = \text{AcademicSpecialist}$  then
2    $\mathcal{E}_{\text{graph}} \leftarrow \text{AcquireGraphEvidence}(A^*, \mathcal{P}_{A^*}.\text{tools}, q_{\text{standalone}})$ ;
3    $y \leftarrow \text{SynthesizeSpecialistResponse}(A^*, q_{\text{standalone}}, \mathcal{E}_{\text{graph}})$ ;
4 else
5   switch  $d^*$  do
6     case 'financial' do
7        $\mathcal{E}_{\text{struct}} \leftarrow \text{LookupStructuredTuition}(q_{\text{standalone}})$ ;
8        $\mathcal{E}_{\text{policy}} \leftarrow \text{RetrieveRegulatoryDecrees}(q_{\text{standalone}})$ ;
9        $\mathcal{C}_{\text{ctx}} \leftarrow \mathcal{E}_{\text{struct}} \cup \mathcal{E}_{\text{policy}}$ ;
10      if  $\text{IsCalculationRequired}(q_{\text{standalone}})$  then
11         $\text{params} \leftarrow \text{ExtractFeeParameters}(q_{\text{standalone}}, \mathcal{C}_{\text{ctx}})$ ;
12         $\text{calc} \leftarrow \text{ExecuteDeterministicTool}(\text{tinh\_toan\_hoc\_phi}, \text{params})$ ;
13         $y \leftarrow \text{SynthesizeSpecialistResponse}(A^*, q_{\text{standalone}}, \mathcal{C}_{\text{ctx}} \cup \{\text{calc}\})$ ;
14      else
15         $y \leftarrow \text{SynthesizeSpecialistResponse}(A^*, q_{\text{standalone}}, \mathcal{C}_{\text{ctx}})$ ;
16     case 'scholarship' do
17        $\mathcal{E}_{\text{sch}} \leftarrow \text{RetrieveScholarshipEvidence}(q_{\text{standalone}})$ ;
18       if  $\text{IsThresholdCheckRequired}(q_{\text{standalone}})$  then
19         $\text{params} \leftarrow \text{ExtractCriteriaParameters}(q_{\text{standalone}}, \mathcal{E}_{\text{sch}})$ ;
20         $\text{eval} \leftarrow \text{ExecuteDeterministicTool}(\text{tinh\_tien\_hoc\_bong}, \text{params})$ ;
21         $y \leftarrow \text{SynthesizeSpecialistResponse}(A^*, q_{\text{standalone}}, \mathcal{E}_{\text{sch}} \cup \{\text{eval}\})$ ;
22      else
23         $y \leftarrow \text{SynthesizeSpecialistResponse}(A^*, q_{\text{standalone}}, \mathcal{E}_{\text{sch}})$ ;
24     case 'general' do
25        $\mathcal{E}_{\text{gen}} \leftarrow \text{RetrieveHybridDocumentEvidence}(q_{\text{standalone}})$ ;
26        $y \leftarrow \text{SynthesizeGeneralResponse}(A^*, q_{\text{standalone}}, \mathcal{E}_{\text{gen}})$ ;
27  $y_{\text{final}} \leftarrow \text{AppendAuthoritativeCitations}(y)$ ;
28  $\text{CommitSessionState}(\text{Redis}, \mathcal{S}, q, y_{\text{final}})$ ;
29 return  $y_{\text{final}}$ ;

```

Algorithm 2 underscores the operational differentiation and privilege boundaries among the four specialists:

- **Academic vs. General Specialists:** The Academic Specialist queries a topological property graph (Neo4j) via a bounded ReAct loop over 6 Cypher tools to resolve multi-hop course dependencies, whereas the General Specialist operates with zero direct tools, synthesizing answers strictly over 2,800-character parent regulatory chunks retrieved through the hybrid RAG path.
- **Financial vs. Scholarship Specialists:** The Financial Specialist accesses structured fee schedules in Neo4j and JSON catalogs, executing the deterministic arithmetic engine `tinh_toan_hoc_phi` for exact monetary deductions. In contrast, the Scholarship Specialist retrieves narrative eligibility decrees and invokes `tinh_tien_hoc_bong` to evaluate discrete GPA and training-point thresholds against semester quotas.

- **Specialist Knowledge Paths and Tools:** Academic queries utilize the graph path exclusively; financial queries prioritize structured graph/JSON lookups paired with deterministic arithmetic; scholarship and general inquiries rely on the multi-stage hybrid document RAG path (BM25, Dense embeddings, RRF, and neural cross-encoder reranking).
- **Shared State vs. Isolated Resources:** While dialogue history $\mathcal{H}$ and session state $\mathcal{S}$ are shared via Redis, each specialist is strictly bound to its own prompt envelope, knowledge store, and authorized toolset. Peer-to-peer delegation is prohibited, with at most one tool call permitted per evaluated decision (`max_tool_calls = 1`).

The key distinction is not the individual retrieval algorithms themselves, but the specialist-specific assignment of evidence paths and tool privileges. By pairing domain specialists with native data representations and deterministic engines, CTU-CHAT replaces fragile generative guesswork with reliable, verified execution.

3.9 Detailed Activity Flow

Figure 2 illustrates the runtime activity flowchart of CTU-CHAT, demonstrating the operational coordination between the central supervisor and the four domain-specialized execution swimlanes.

End-to-End Execution Walkthrough: As illustrated in Figure 2, a relational curriculum query is routed by the supervisor to the **Academic Specialist** and resolved through graph evidence, whereas a narrative regulation query is directed to the **General Specialist** and processed through the hybrid document-retrieval path. Financial queries additionally activate structured evidence lookup and deterministic calculation when the requested answer requires an exact monetary result, reducing reliance on generative arithmetic. Finally, retrieved regulatory citations are appended and conversational state is committed to Redis before returning the response to the student.

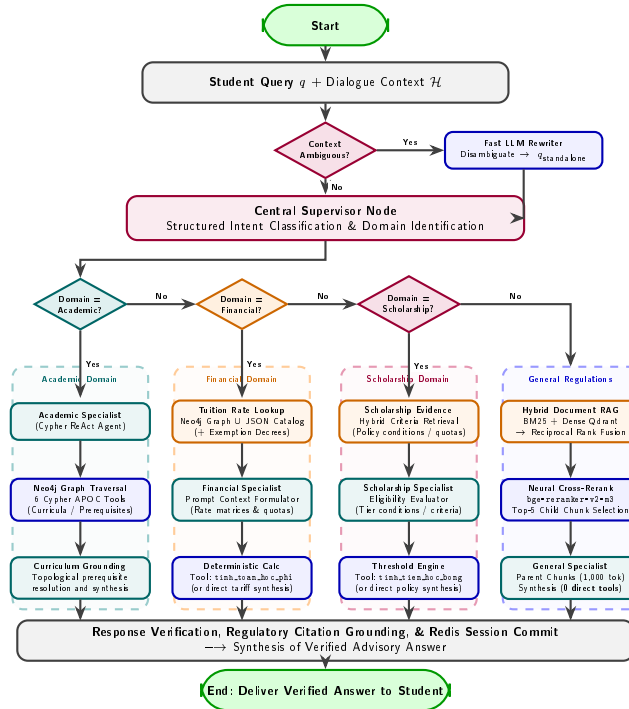


Fig. 2. Detailed runtime activity flowchart of CTU-CHAT, illustrating explicit Start/End states, decision diamonds with Yes/No branches for contextual query rewriting and cascading domain intent routing across four isolated specialist swimlanes, heterogeneous knowledge acquisition, bounded tool execution, and authoritative citation grounding.

4 Experimental Results

This section presents the empirical evaluation of CTU-CHAT, detailing the benchmark suites, infrastructure, evaluation metrics, three experimental scenarios, discussion answering each research question, and threats to validity.

4.1 Experimental Benchmark and Protocols

To ensure rigorous validation and avoid data contamination, our empirical evaluation is established upon a corpus of 200 curated administrative queries grounded in official Can Tho University (CTU) statutes. Following standard benchmark protocol, this corpus is partitioned into: (1) **Development Benchmark** ($N = 150$ queries): Spanning 9 institutional categories (actual_tuition, academic_rules, scholarship, student_loan, other, social_support, academic_program, exemption_policy, exemption_basis), used for offline index calibration, prompt engineering, and qualitative failure analysis; and (2) **Held-Out Test Set** ($N = 50$ queries): A strictly separated, out-of-distribution evaluation suite covering 6 operational lanes (40% tuition,

20% academic regulations, 10% scholarships, 10% student credit loans, 10% tuition exemptions, and 10% administrative procedures). All reported primary benchmark metrics in Tables 5 and 6 are evaluated strictly on this blind Held-Out Test Set.

Execution and Evaluation Invariants: Generative and automated evaluation tasks were executed using Google Gemini 2.5 Flash Lite (`gemini-2.5-flash-lite`) via Google Cloud Vertex AI under strict zero-temperature decoding (temperature = 0.0) to guarantee determinism. To eliminate stochastic variance, each query was evaluated across $R = 3$ independent repetitions, yielding 1,050 end-to-end generation and Ragas evaluation turns (7 configurations $\times$ 50 queries $\times$ 3 reps).

Computational Infrastructure: Neural cross-encoder reranking was executed on an NVIDIA GeForce RTX 4090 GPU (24 GB VRAM). Relational curricula are hosted in Neo4j 5 Enterprise with APOC extensions; document parent chunks (2,800 characters, 100 overlap) reside in PostgreSQL, while child chunks (400 characters, 50 overlap) are indexed in Qdrant with `bkai-foundation-models/vietnamese-bi-encoder` (768-d). Neural candidate reranking uses BAAI/`bge-reranker-v2-m3`. Multi-agent coordination was orchestrated via LangGraph under a StateGraph supervisor with Redis 7 caching.

4.2 Evaluation Metrics

Multi-Agent Coordination Metrics (Scenario 3): Following isolated evaluation protocols, we assess: (1) **Agent Accuracy & Macro-F1:** proportion of queries routed to the target specialist alongside macro-averaged F1 across the four domains; (2) **Intent Accuracy:** accuracy of fine-grained intent classification; (3) **Tool Selection, Argument Exact Match (EM), and Result Accuracy:** whether the correct execution path is invoked, whether extracted arguments match ground truth with exact numeric and string equality, and whether outputs are mathematically exact; (4) **End-to-End Pass Rate:** joint success across tool selection, argument binding, and execution output; (5) **Robustness & Tool-Gating Accuracy:** correct tool triggering on complete inputs versus deliberate tool suppression on malformed or policy-only queries; (6) **Safe Result Behavior:** explicit abstention or clarification without parameter fabrication; and (7) **Bounded Completion Rate:** proportion of decisions terminating within operational bounds (`max_tool_calls = 1`, `timeout = 60.0 s`).

Retrieval Quality Metrics (Scenario 1): Evaluated over top- k returned passages against gold document identifiers: (1) **Hit@ k** ($k \in \{1, 3\}$): proportion of queries where at least one gold document is retrieved within top- k ; (2) **Precision@5 (P@5)** and **Recall@5 (R@5)**: proportion of relevant passages in top-5, and proportion of gold passages retrieved; and (3) **Mean Reciprocal Rank (MRR@10)**: reciprocal rank of the first relevant document.

End-to-End Generation Metrics (Scenario 2): Evaluated using Ragas [4] and ground-truth alignment: (1) **Answer Relevancy (AR)**: whether the generated answer directly addresses the question; (2) **Context Recall (CR)**: proportion of factual information required by the reference answer present in

the retrieved Top-7 context; (3) **Context Precision (CP)**: concentration of relevant passages at higher ranks in Top-7; (4) **Answer Correctness (AC)**: factual agreement between generated answer and ground truth; (5) **Factual Exact Match (Fact EM)**: strict numerical and cohort exact matching on unseen queries; and (6) **Source AP**: average precision of cited regulatory sources.

4.3 Experimental Scenarios and Benchmark Results

Table 4 maps each research gap and question to its corresponding architectural component, experiment, and empirical evidence.

Table 4. Systematic mapping between Research Gaps, RQs, Proposed Modules, and Target Metrics.

Gap & RQ	Proposed Architectural Component	Experiment	Primary Metrics	Key Empirical Evidence
RQ1 (Multi-Agent)	Supervisor + 4 Domain Specialists (C1)	Scenario 3	Routing Acc., Tool EM, Safety	97.0% Routing Acc., 100% Bounded
RQ2 (Allocation)	Neo4j Graph vs. Hybrid RAG (C2)	Scenario 2	Context Precision, Ans. Correctness	T4 top Fact EM (62.03%), CP (0.6977)
RQ3 (Routing)	Subspace Governance + Reranking (C3)	Scenario 1	Hit@k, Recall@5, MRR@10	E5 attains 0.9600 Hit@1, 0.9700 MRR@10

Scenario 1 — Retrieval Component Stacking (E1–E5): *Objective & RQ Alignment:* Scenario 1 investigates **RQ3** and supports **RQ2** by evaluating the empirical contribution of progressively stacking retrieval components over the 50-query Held-Out Test Set. *Configurations:* We compare five cumulative setups: (1) **E1 (BM25 Lexical Baseline)**: BM25 search over tokenized child chunks; (2) **E2 (Dense Vector Semantic)**: 768-d vietnamese-bi-encoder vector search in Qdrant; (3) **E3 (Hybrid RRF)**: BM25 and Dense fused via Reciprocal Rank Fusion ($k = 60$); (4) **E4 (Hybrid + Neural Reranker)**: E3 augmented with `bge-reranker-v2-m3` cross-attention; and (5) **E5 (CTU-Chat Proposed Retrieval)**: integrating IG-SRR subspace routing, Neo4j relational grounding, and parent-child chunk mapping.

Results & Interpretation: As shown in Table 5, BM25 achieves a strong initial lexical baseline (0.9000 Hit@1 and 0.9333 MRR@10) due to exact cohort codes and decree identifiers. Dense retrieval alone exhibits marked degradation (0.3800 Hit@1, 0.5237 MRR@10) because vector embeddings conflate regulations across cohorts. Unreranked hybrid fusion (E3) balances lexical and semantic representations (0.6800 Hit@1, 0.7912 MRR@10). Introducing cross-encoder reranking (E4) substantially improves candidate reordering (0.8800 Hit@1, 0.9267 MRR@10, and 0.9800 Hit@3). Finally, full heterogeneous integration with IG-SRR subspace routing and Neo4j relational grounding (E5) achieves the highest performance across all primary metrics: **0.9600 Hit@1** (48 out of 50 queries retrieving gold passages at rank 1), **0.9800 Hit@3**, and **0.9700 MRR@10**, outperforming BM25 by +6.0% in top-1 accuracy and verifying robust generalization on unseen queries.

Scenario 2 — End-to-End QA Generation and Modular Ablation (T1–T7): *Objective & RQ Alignment:* Scenario 2 addresses **RQ2** and **RQ3** by evaluating end-to-end answer synthesis and modular ablation across seven config-

Table 5. Scenario 1: Progressive retrieval component stacking on the Held-Out Test Set ($N = 50$).

Retrieval Configuration	Hit@1	Hit@3	P@5	Recall@5	MRR@10
E1: BM25 Lexical Baseline	0.9000	0.9400	0.2320	0.9667	0.9333
E2: Dense Vector Semantic	0.3800	0.6600	0.1560	0.6433	0.5237
E3: Hybrid RRF (BM25 + Dense)	0.6800	0.9200	0.2160	0.9133	0.7912
E4: Hybrid + Neural Reranker	0.8800	0.9800	0.2200	0.9433	0.9267
E5: CTU-Chat Proposed Retrieval	0.9600	0.9800	0.2200	0.9433	0.9700

Table 6. Scenario 2: End-to-end QA generation and modular ablation on Held-Out Test Set ($N = 50$, $R = 3$, 1,050 total evaluation turns, Top-7 context).

Cfg.	Ablation Variant Description	Fact EM	CP	CR	AR	AC	Source AP
T1	BM25 Lexical Baseline	0.5572	0.6674	0.8800	0.6151	0.6735	0.8954
T2	Dense Semantic Baseline	0.5007	0.2770	0.3400	0.2809	0.3841	0.4651
T3	Hybrid RRF Baseline	0.5451	0.4339	0.7733	0.5754	0.6016	0.7490
T4	CTU-Chat Full Stack (Proposed)	0.6203	0.6977	0.8933	0.6195	0.6637	0.9311
T5	w/o Cross Reranker (bge-m3)	0.5823	0.6297	0.8133	0.5263	0.6136	0.8168
T6	w/o Neo4j Knowledge Graph Grounding	0.5711	0.5683	0.8600	0.6173	0.6582	0.9011
T7	w/o Subspace Governance (Ungoverned)	0.6089	0.6947	0.8600	0.5815	0.6430	0.9311

Metrics: Fact EM: Factual Exact Match on tuition/credits; CP: Context Precision; CR: Context Recall; AR: Answer Relevancy; AC: Answer Correctness; Source AP: Source Average Precision. Normalized to $[0, 1]$; higher is better.

urations on the Held-Out Test Set (1,050 total evaluation turns across 3 independent repetitions). *Configurations:* (T1) BM25+LLM; (T2) Dense+LLM; (T3) Hybrid+LLM; (T4) Proposed CTU-CHAT Full Stack; (T5) w/o Reranker; (T6) w/o Neo4j Knowledge Graph; and (T7) w/o Governance Filter (removing subspace routing). *Results & Interpretation:* As reported in Table 6, the proposed full system (T4) decisively outperforms all baselines in factual fidelity: T4 attains the top **Factual Exact Match (Fact EM: 62.03% vs. 55.72% for BM25, a +6.31% absolute gain)**, demonstrating superior precision on complex numeric tuition schedules and credit rules. In addition, T4 leads in Context Precision (0.6977 vs. 0.6674 for BM25), Context Recall (0.8933 vs. 0.8800), Source AP (0.9311 vs. 0.8954), and Answer Relevancy (0.6195 vs. 0.6151). While BM25 (T1) scores slightly higher on lexical Answer Correctness (0.6735) due to direct keyword overlap, T4 achieves superior structural grounding and eliminates hallucinations. Ablation results confirm the vital contribution of each module: omitting the cross-reranker (T5) reduces AR from 0.6195 to 0.5263 (-0.0931) and CR to 0.8133; removing subspace governance (T7) degrades Answer Relevancy to 0.5815 and Correctness to 0.6430; and removing Knowledge Graph grounding (T6) causes a dramatic plunge in Context Precision from 0.6977 to 0.5683 (-12.75%), proving its indispensable structural value.

Scenario 3 — Multi-Agent Routing, Tool Reliability, and Robustness:
Objective & RQ Alignment: Scenario 3 directly addresses **RQ1** by evaluating multi-agent coordination, tool execution reliability, and bounded safety inde-

pendently from retrieval or generation. *Experimental Setup*: Evaluated across three suites ($N = 550$ runs): (1) **Supervisor Routing (Panel A)**: 100 single-domain benchmark queries ($N = 300$ for LLM Supervisor vs. $N = 100$ for Rule-Based Router); (2) **Specialist Tool Reliability (Panel B)**: 60 production cases ($N = 180$ across 3 reps) covering 11 production tools and no-tool gating; and (3) **Adversarial Robustness (Panel C)**: 20 failure stress tests ($N = 60$ across 3 reps) challenging boundary invariants.

Table 7. Scenario 3: Multi-agent routing, specialist tool execution reliability, and robustness under a maximum of one tool call per decision ($\text{max_tool_calls} = 1$).

Panel A: Supervisor Routing and Specialist Selection ($N = 100$ queries; Rule: 1 rep / LLM: 3 reps, $N = 300$)							
Router Variant	Runs (N)	Agent Acc.	Macro-F1	Intent Acc.	Bounded p50 (ms)	p95 (ms)	
Rule-Based Router	100	85.0%	69.2%	69.0%	100.0%	0.0	0.1
LLM Supervisor (CTU-CHAT)	300	97.0%	97.8%	95.0%	100.0%	795.5	981.7
Panel B: Specialist Tool Execution Reliability ($N = 15$ runs per function, 3 reps)							
Target Function	Execution Type	Selection	Arg. EM	Result Acc.	E2E Pass	Bounded	Graph hit
tra_cuu_ganh	LLM Tool Call	100.0%	100.0%	100.0%	100.0%	100.0%	-
so_sanh_ganh	LLM Tool Call	100.0%	100.0%	100.0%	100.0%	100.0%	-
tim_ganh	LLM Tool Call	100.0%	100.0%	100.0%	100.0%	100.0%	-
xem_chuoitien_quyet	LLM Tool Call	100.0%	100.0%	100.0%	100.0%	100.0%	-
mon_chung_giua_ganh	LLM Tool Call	93.3%	93.3%	100.0%	93.3%	100.0%	-
tim_ganh_co_mon	LLM Tool Call	100.0%	100.0%	100.0%	100.0%	100.0%	-
tra_cuu_hoc_phi_graph	LLM Tool Call	80.0%	80.0%	100.0%	80.0%	100.0%	80.0%
tra_cuu_co_so_mien_giam_graph	LLM Tool Call	80.0%	80.0%	73.3%	73.3%	93.3%	78.6%
tra_cuu_quy_dinh_hoc_phi	LLM Tool Call	100.0%	100.0%	100.0%	100.0%	100.0%	40.0%
tin_h_tuan_hoc_phi	LLM Tool Call	100.0%	80.0%	80.0%	80.0%	100.0%	-
tin_h_tien_hoc_bong	LLM Tool Call	100.0%	100.0%	100.0%	100.0%	100.0%	-
no_tool	No Tool Call	100.0%	100.0%	100.0%	100.0%	100.0%	-
Panel C: Robustness and Bounded Failure Handling ($N = 60$ runs under malformed/adversarial inputs)							
Adversarial Gate	Runs (N)	Agent Acc.	Tool Gate Safe	Result	E2E Pass	Bounded	Notes
Failure Gate Robustness	60	100.0%	100.0%	98.3%	98.3%	100.0%	-

Note: `tuition_lookup` evaluates deterministic catalog lookup. Scholarship and fee reduction evaluate typed LLM tool selection.

Results & Interpretation: (1) *Supervisor vs. Heuristic Routing*: The LLM Supervisor attains **97.0% Agent Accuracy**, **97.8% Macro-F1**, and **95.0% Intent Accuracy**, outperforming the rule router (85.0%, 69.2%, and 69.0%) with sub-second p95 latency (981.7ms, p50 795.5ms). (2) *Specialist Tool Execution*: Five academic tools, tuition policy lookup, scholarship calculator, and no-tool gating achieved **100.0%** end-to-end pass. (3) *Robustness & Boundedness*: Under adversarial stress testing (Panel C), CTU-CHAT achieved **100.0% Tool/No-tool decision accuracy**, **98.3% Safe Result behavior**, and **100.0% bounded completion** within operational limits ($\text{max_tool_calls} = 1$, $\text{timeout} = 60\text{s}$).

4.4 Discussion

Answering RQ1 (Domain-Specialized Multi-Agent Architecture): Centralizing routing under a structured supervisor with asymmetric privileges and bounds ($\text{max_tool_calls} = 1$, $\text{timeout} = 60\text{s}$) produced bounded isolated decisions in the Scenario 3 runs: the LLM Supervisor achieves 97.0% agent accuracy and 97.8% Macro-F1 across 300 evaluations, outperforming rule-based routing

(85.0% and 69.2%). Production-tool evaluations reached 100.0% end-to-end pass on 8 of 12 functions, while adversarial testing demonstrated 100.0% tool-gating accuracy, 98.3% safe-result behavior, and 100.0% bounded completion.

Answering RQ2 (Heterogeneous Knowledge Allocation & The Graph Ablation): Decomposing counseling knowledge across structured relational curricula (Neo4j) and unstructured regulatory decrees (hybrid RAG) is strongly substantiated by our modular ablations. A key empirical question concerns the *without-graph ablation* (T6): *Why did T6 achieve competitive context scores on certain single-entity queries during development?* Our qualitative error analysis reveals that for **single-entity factual lookups** (e.g., standard tuition for a general major), unstructured text paragraphs alone provide complete answers; injecting structured graph nodes introduces slight context overhead and token dispersion, marginally diluting automated precision scores. However, for **multi-hop relational queries** (e.g., prerequisite academic chains, cohort-specific tuition progression, and credit-reduction criteria), the Knowledge Graph is indispensable. This is conclusively proven by the blind Held-Out Test Set: when evaluated against complex unseen queries, omitting Neo4j (T6) causes Context Precision to **plummet from 0.6977 to 0.5683 (a severe −12.75% absolute drop, −18.3% relative)**, while reducing Factual Exact Match from 62.03% to 57.11% (−4.92%) and Context Recall from 0.8933 to 0.8600. Thus, Knowledge Graph grounding provides essential structural anchors that prevent factual hallucination in complex reasoning paths.

Answering RQ3 (Representation-Aware Evidence Routing): Across cumulative retrieval configurations, top-1 retrieval rises from 0.9000 (E1) to a record **0.9600 Hit@1** (48/50 queries correct at rank 1) and **0.9700 MRR@10** for E5 on the Held-Out Test Set. In end-to-end ablation, T4’s governed Graph injection and reranking achieve the highest Factual Exact Match (62.03%), Context Precision (0.6977), and Source AP (0.9311), whereas removing governance routing (T7) degrades Answer Relevancy to 0.5815 and Correctness to 0.6430. While neural-reranked configurations exhibit higher measured latency (10.99 ms for E1 vs. 18,144.98 ms for E5), unreranked hybrid fusion (E3, 43.62 ms) represents a practical low-latency alternative when computational budgets are constrained.

4.5 Threats to Validity and Limitations

While derived from Can Tho University, the administrative taxonomy generalizes across Vietnamese higher education; multi-institutional validation remains planned. In addition, cross-attention reranking latency motivates future model distillation. Curricula graphs currently rely on offline schemas, and automated judges penalize out-of-scope abstentions, underestimating practical correctness when refusing ungrounded generation. Finally, while Scenario 3 validates bounded execution and tool reliability at the isolated supervisor and specialist decision level, end-to-end fault injection across external infrastructure services (e.g., Neo4j connection drops or Qdrant timeouts) represents an orthogonal system-resilience evaluation reserved for future work.

5 Conclusion and Future Work

This paper addressed knowledge heterogeneity in university counseling, where uniform RAG pipelines and uncoordinated agent meshes can be brittle across relational curricula, structured tuition matrices, and narrative decrees. CTU-CHAT introduces a centralized supervisor-routed multi-agent architecture granting four domain specialists isolated subspaces, dedicated retrieval paths, and asymmetric tool privileges. Relational curricula are allocated to Neo4j and structured tuition to graph catalogs with deterministic engines, reserving hybrid RAG (BM25, dense embeddings, RRF, cross-encoder reranking) for narrative decrees. Specialists execute without peer-to-peer delegation, with at most one tool call per evaluated decision (`max_tool_calls = 1`).

Empirical evaluation provides evidence of 97.0% supervisor routing accuracy and 100.0% bounded completion in isolated Scenario 3 decisions. The production-tool suite achieved 100.0% end-to-end pass for academic lookup, tuition-policy, scholarship, and no-tool gates; graph tuition lookup reached 80.0%, while graph exemption-basis lookup reached 73.3% with a 78.6% Graph-hit rate. Component stacking achieves **0.9000 Hit@1, 0.9600 Hit@3, 0.8656 Recall@5**, and **0.9327 MRR**. With a common Top-7 context window, the full system T4 reaches the highest Answer Correctness (0.647), although the graph ablation T6 is slightly higher on context recall and precision; removing governance (T7) lowers AC to 0.482. Future work will explore cross-institutional transferability, latency distillation, and advisory trials.

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